Online Appendix – Not for Print

Smoke Signals: Understanding Temporal Dynamics of Wildfire Exposure on Health and Education

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Appendix Figures and Tables

Table A1: Number of wildfires reported in Chile by season and burned area range

Season	$<$ 1 ha	1-10 ha	10-100 ha	100-200 ha	200-500 ha	500-1000 ha	$>$ 1000 ha
2004-2005	4457	1766	343	38	28	10	11
2005-2006	3831	1312	218	16	14	4	1
2006-2007	3723	1205	167	16	13	10	10
2007-2008	4706	1869	327	36	26	6	3
2008-2009	4099	1597	369	32	37	11	12
2009-2010	2763	1016	202	25	38	12	13
2010-2011	3339	1311	234	31	24	5	7
2011-2012	3789	1368	285	31	22	6	8
2012-2013	3941	1435	248	15	11	1	0
2013-2014	4246	1615	364	44	37	15	19
2014-2015	5100	2173	617	72	66	28	17
2015-2016	4466	1829	426	27	24	5	7
2016-2017	3178	1490	419	40	58	32	57
2017-2018	3814	1729	468	40	22	6	2
2018-2019	4726	1899	450	51	29	14	14
2019-2020	5274	2127	568	72	48	21	17
2020-2021	5120	1614	311	28	20	5	3
2021-2022	4788	1571	445	45	53	17	28

Table A2: Descriptive statistics of selected wildfires by season (only fires with area affected greater than 200 ha)

Season	N	Mean	Median	Std	Min	Max	P25	P75	Total
2002-2003	34	648.31	345.00	765.16	220.00	3583.00	280.00	676.25	22042.57
2003-2004	48	620.90	424.50	524.18	206.00	2200.00	278.50	657.78	29803.13
2004-2005	45	613.33	420.00	480.01	203.80	2154.00	255.00	800.00	27600.00
2005-2006	14	422.84	325.00	304.07	201.00	1300.00	223.25	465.50	5919.69
2006-2007	32	1025.77	742.23	917.33	200.00	5046.00	450.00	1498.00	32824.10
2007-2008	30	620.08	382.76	855.81	208.50	3740.00	277.50	461.75	18602.31
2008-2009	56	695.15	412.50	788.68	201.00	3975.91	253.75	662.82	38928.58
2009-2010	59	749.44	400.00	808.72	201.00	3500.00	250.00	840.00	44216.96
2010-2011	36	701.43	375.00	811.03	210.00	3891.60	268.75	703.00	25251.47
2011-2012	4	7469.73	2380.13	11757.05	223.00	24895.65	468.25	9381.60	29878.90
2012-2013	12	296.35	270.00	104.14	203.00	523.40	206.75	348.10	3556.20
2013-2014	69	1209.64	482.00	2432.43	202.00	14805.00	300.00	1050.00	83465.09
2014-2015	111	830.21	395.00	1593.80	200.49	13833.00	268.50	618.96	92153.14
2015-2016	35	573.48	353.00	491.89	205.00	2082.10	233.75	746.00	20071.80
2016-2017	144	3792.21	689.50	14402.29	203.00	159812.58	339.50	2280.45	546077.82
2017-2018	28	428.57	296.22	357.18	200.50	2069.00	262.70	478.72	12000.04
2018-2019	56	950.21	504.82	1999.83	207.00	15145.00	302.18	949.95	53211.61
2019-2020	85	796.54	439.20	1642.91	208.96	14987.90	273.76	826.52	67706.08
2020-2021	27	615.65	311.43	751.68	202.89	3420.03	256.79	577.08	16622.42
2021-2022	96	1034.47	445.85	1693.76	203.17	13768.00	253.72	1066.49	99309.45

Note: This table reports descriptive statistics by wildfire season, considering only wildfires with an affected area greater than 200 hectares. Only the wildfires included in the atmospheric trajectory modeling using HYSPLIT are reported here.

Table A3: Effects of Wildfire Exposure on Students' Attendance

	Attendance	
	(1)	(2)
Wildfire Exposure	-0.4312	-0.4187
	(0.507)	(0.591)
Number of students	389,492	389,492
Mean of Dep. Var.	88.846	88.846
Observations	3,894,920	3,894,920
R-Squared	0.47	0.47
Fixed Effects		
Student	Y	Y
Month	Y	
Month×Region		Y

Notes: Each observation correspond to a student in all grades in 2013 from regions V and VII, which experienced wildfire exposure in June. Within region, the exposure differs between municipalities, and therefore students. The independent variable measures exposure to wildfire smoke plumes in 8-hour intervals. To interpret the estimated coefficients as the effect of an additional full day of exposure (i.e., 24 hours), the coefficients should be multiplied by 3. Standard errors clustered at the school's municipality level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Heterogeneous Effects of In-Utero Exposure to Wildfire on Birth Outcomes, by Mother's Education Level

	Birth Weight			Birth Size			Gestational Weeks		
Mother's education level:	College	Secondary	Primary	College	Secondary	Primary	College	Secondary	Primary
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Smoke Exposure IU [1rd trimester]	3.481***	-0.438	0.960	0.011**	0.002	0.001	0.006**	-0.003	-0.004
(8-hours period)	(0.769)	(1.358)	(2.931)	(0.005)	(0.010)	(0.016)	(0.002)	(0.003)	(0.009)
Smoke Exposure IU [2rd trimester]	-2.200	-0.625	-7.125***	-0.012***	-0.012***	-0.039***	-0.007**	-0.008***	-0.032***
(8-hours period)	(1.352)	(0.683)	(2.601)	(0.004)	(0.004)	(0.012)	(0.003)	(0.003)	(0.010)
Smoke Exposure IU [3rd trimester]	0.654	-1.610	3.317**	0.002	-0.002	0.010	-0.002	-0.005	0.001
(8-hours period)	(0.868)	(1.197)	(1.586)	(0.005)	(0.007)	(0.010)	(0.005)	(0.004)	(0.008)
Mean of Dep. Var.	3273.625	3332.116	3349.322	49.105	49.334	49.362	38.283	38.532	38.624
Observations	1,038,390	1,872,869	419,041	1,038,390	1,872,869	419,041	1,038,390	1,872,869	419,041
R-Squared	0.01	0.01	0.01	0.02	0.02	0.02	0.01	0.01	0.01
FDR-adjusted p-values (β_1)	0.001	1.000	1.000	0.203	1.000	1.000	0.203	1.000	1.000
FDR-adjusted p-values (β_2)	0.173	0.365	0.022	0.022	0.012	0.012	0.026	0.022	0.012
FDR-adjusted p-values (β_3)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Fixed Effects									
Municipality	Y	Y	Y	Y	Y	Y	Y	Y	Y
Calendar Month-of-birth×Year-of-birth	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year-of-birth×Season×Region	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Sample consists of all births (individual measures) between 2005-2018. Trimesters of pregnancy are calculated since conception, such that the first two trimesters are thirteen weeks each, whereas the third trimester may vary according to the actual week of birth. The independent variable measures exposure to wildfire smoke plumes in 8-hour intervals. To interpret the estimated coefficients as the effect of an additional full day of exposure (i.e., 24 hours), the coefficients should be multiplied by 3 Standard errors clustered at the 0.75°× 0.75° spatial grid (roughly, 70×70 Km) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A5: Heterogeneous Effects of In-Utero Exposure to Wildfire on Lifetime Hospitalizations, by Mother's Education Level

	Infants ($\leq$ 1 years old)			Toddlers (1-3 years old)			All ages		
Mother's education level:	College	Secondary	Primary	College	Secondary	Primary	College	Secondary	Primary
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Smoke Exposure IU [1rd trimester]	0.133	0.232	0.391	0.270	-0.112	-0.087	0.488	-0.080	-0.018
(8-hours period)	(0.218)	(0.151)	(0.609)	(0.486)	(0.109)	(0.222)	(0.551)	(0.205)	(0.702)
Smoke Exposure IU [2rd trimester]	-0.024	0.385	0.354	-0.050	0.028	0.250	-0.104	0.244	0.750
(8-hours period)	(0.214)	(0.317)	(0.388)	(0.182)	(0.237)	(0.286)	(0.221)	(0.566)	(0.492)
Smoke Exposure IU [3rd trimester]	-0.077	0.271	0.509	0.122	0.134	-0.252	0.125	0.146	0.381
(8-hours period)	(0.196)	(0.282)	(0.336)	(0.199)	(0.166)	(0.230)	(0.310)	(0.401)	(0.426)
Mean of Dep. Var.	33.160	41.470	52.310	21.651	22.404	25.323	64.778	74.361	89.290
Observations	1,038,390	1,872,869	419,041	1,038,390	1,872,869	419,041	1,038,390	1,872,869	419,041
R-Squared	0.01	0.01	0.02	0.01	0.00	0.01	0.01	0.01	0.02
FDR-adjusted p-values (β_1)	1.000	0.905	1.000	1.000	1.000	1.000	1.000	1.000	1.000
FDR-adjusted p-values (β_2)	0.698	0.318	0.365	0.698	0.698	0.365	0.668	0.668	0.190
FDR-adjusted p-values (β_3)	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Fixed Effects									
Municipality	Y	Y	Y	Y	Y	Y	Y	Y	Y
Calendar Month-of-birth×Year-of-birth	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year-of-birth×Season×Region	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes: Sample consists of all births (individual measures) between 2005-2018. Trimesters of pregnancy are calculated since conception, such that the first two trimesters are thirteen weeks each, whereas the third trimester may vary according to the actual week of birth. The independent variable measures exposure to wildfire smoke plumes in 8-hour intervals. To interpret the estimated coefficients as the effect of an additional full day of exposure (i.e., 24 hours), the coefficients should be multiplied by 3. Standard errors clustered at the 0.75°× 0.75° spatial grid (roughly, 70×70 Km) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A6: Effects of In-Utero Exposure to Wildfire on Birth Outcomes

	Birth Weight		Birth Size		Gestational Weeks	
	(1)	(2)	(3)	(4)	(5)	(6)
Smoke Exposure IU [Last two weeks] (8-hours period)	-3.635* (2.097)		-0.019 (0.015)		-0.015** (0.006)	
Smoke Exposure IU [Third trimester] (8-hours period)		-0.655 (0.518)		-0.001 (0.003)		-0.005* (0.003)
Mean of Dep. Var.	3315.984	3315.984	49.266	49.266	38.466	38.466
Observations	3,331,299	3,331,299	3,331,299	3,331,299	3,331,299	3,331,299
R-Squared	0.01	0.01	0.02	0.02	0.01	0.01
FDR-adjusted p-values	0.122	1.000	0.149	1.000	0.101	0.964
Fixed Effects						
Municipality	Y	Y	Y	Y	Y	Y
Month×Year	Y	Y	Y	Y	Y	Y
Year×Season×Region	Y	Y	Y	Y	Y	Y

Notes: Sample consists of all births (individual measures) between 2005-2018. The independent variable measures exposure to wildfire smoke plumes in 8-hour intervals. To interpret the estimated coefficients as the effect of an additional full day of exposure (i.e., 24 hours), the coefficients should be multiplied by 3. Standard errors clustered at the 0.75°× 0.75° spatial grid (roughly, 70×70 Km) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A7: Effects of In-Utero Exposure to Wildfire on Lifetime Hospitalizations

	Infants ($\leq$ 1 years old)		Toddlers (1-3 years old)		All ages	
	(1)	(2)	(3)	(4)	(5)	(6)
Smoke Exposure IU [Last two weeks] (8-hours period)	0.006* (0.003)		0.009* (0.005)		0.005 (0.007)	
Smoke Exposure IU [Third trimester] (8-hours period)		0.001 (0.002)		0.001 (0.002)		0.001 (0.003)
Mean of Dep. Var.	0.402	0.402	0.225	0.225	0.733	0.733
Observations	3,331,299	3,331,299	3,331,299	3,331,299	3,331,299	3,331,299
R-Squared	0.01	0.01	0.00	0.00	0.01	0.01
FDR-adjusted p-values	0.122	1.000	0.122	1.000	0.183	1.000
Fixed Effects						
Municipality	Y	Y	Y	Y	Y	Y
Month×Year	Y	Y	Y	Y	Y	Y
Year×Season×Region	Y	Y	Y	Y	Y	Y

Notes: Sample consists of all births (individual measures) between 2005-2018. The independent variable measures exposure to wildfire smoke plumes in 8-hour intervals. To interpret the estimated coefficients as the effect of an additional full day of exposure (i.e., 24 hours), the coefficients should be multiplied by 3. Standard errors clustered at the 0.75°× 0.75° spatial grid (roughly, 70×70 Km) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

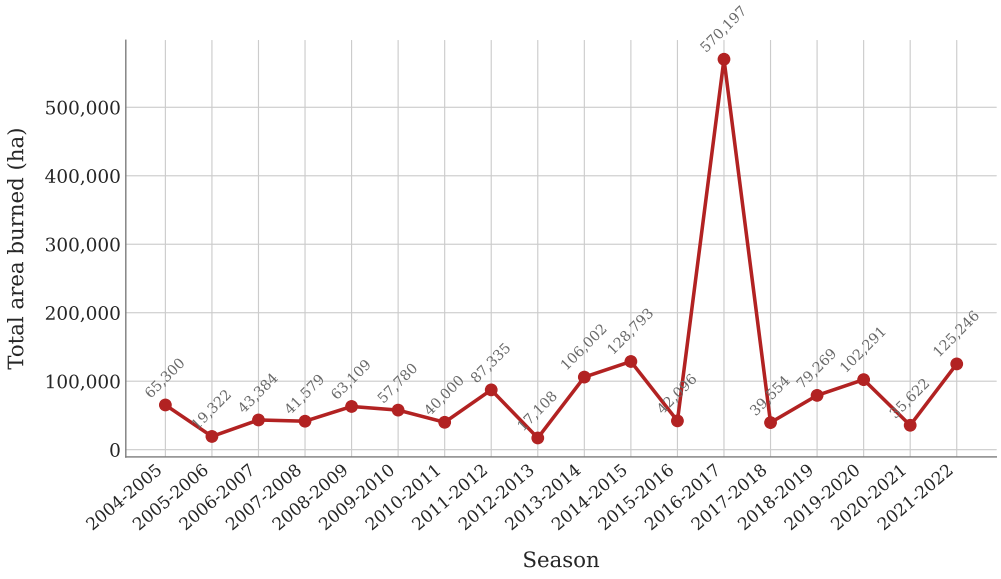
Table A8: Effects of In-Utero Wildfire Exposure on Standardized Test Scores at 4th grade by Different Weeks of Exposure

	Math Test Scores			Verbal Test Scores		
	(1)	(2)	(3)	(4)	(5)	(6)
Smoke Exposure IU [Last two weeks]	-4.48			-7.14		
(8-hours period)	(6.93)			(7.67)		
Smoke Exposure IU [Last Month]		5.22			5.35	
(8-hours period)		(3.78)			(5.67)	
Smoke Exposure IU [Third trimester]			0.98			0.52
(8-hours period)			(1.79)			(1.71)
Mean of Dep. Var.	264.216	264.216	264.216	271.652	271.652	271.652
Observations	620,999	620,999	620,999	618,809	618,809	618,809
R-Squared	0.28	0.28	0.28	0.19	0.19	0.19
FDR-adjusted p-values (X1)	1.000	0.506	1.000	1.000	0.506	1.000
Fixed Effects						
Municipality of birth	Y	Y	Y	Y	Y	Y
Year of test	Y	Y	Y	Y	Y	Y
School	Y	Y	Y	Y	Y	Y
Calendar Month-of-birth × Year-of-birth	Y	Y	Y	Y	Y	Y
Region×Season×Year-of-birth	Y	Y	Y	Y	Y	Y

Notes: Sample consists of students (individual measures) born between 2004 and 2008 taking the exam at 9, 10, or 11 years old. The independent variable measures exposure to wildfire smoke plumes in 8-hour intervals. To interpret the estimated coefficients as the effect of an additional full day of exposure (i.e., 24 hours), the coefficients should be multiplied by 3. Standard errors clustered at the school's municipality level in parentheses.

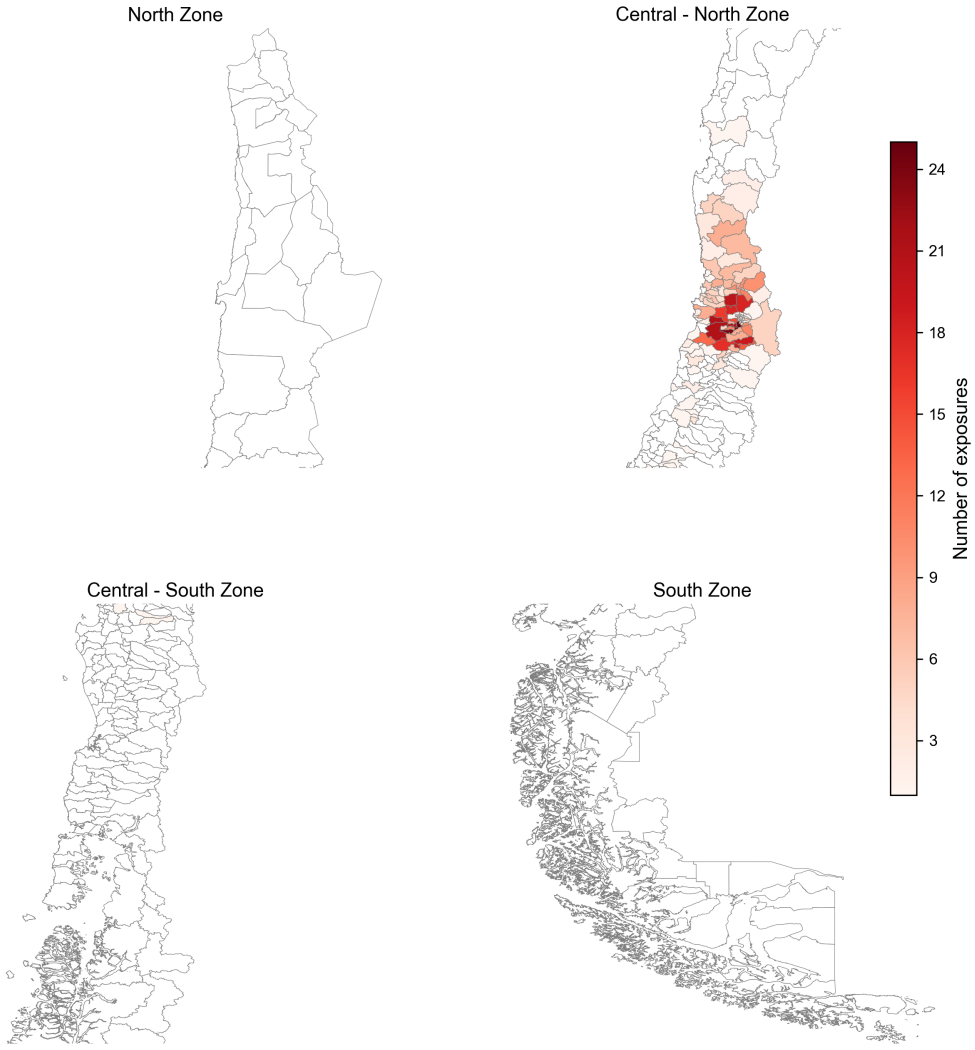
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure A1: Total area burned by wildfires per season in Chile (2004-2022).



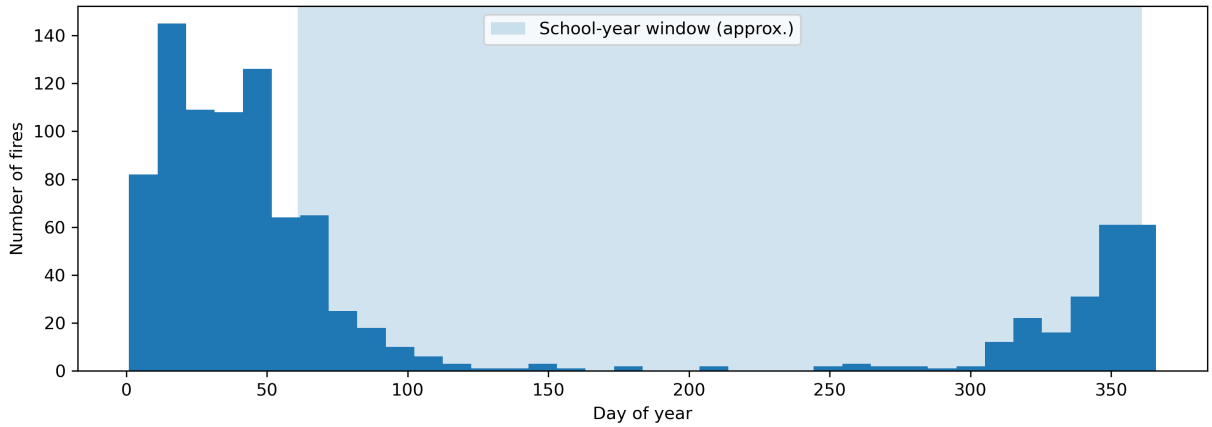
Notes: The figure reports the total area burned (ha) by wildfire season in Chile. Each point corresponds to the national total for the indicated season (e.g., 2016–2017). *Source:* CONAF. *Authors' calculations.*

Figure A2: Wildfire exposure by municipality – season 2016-2017

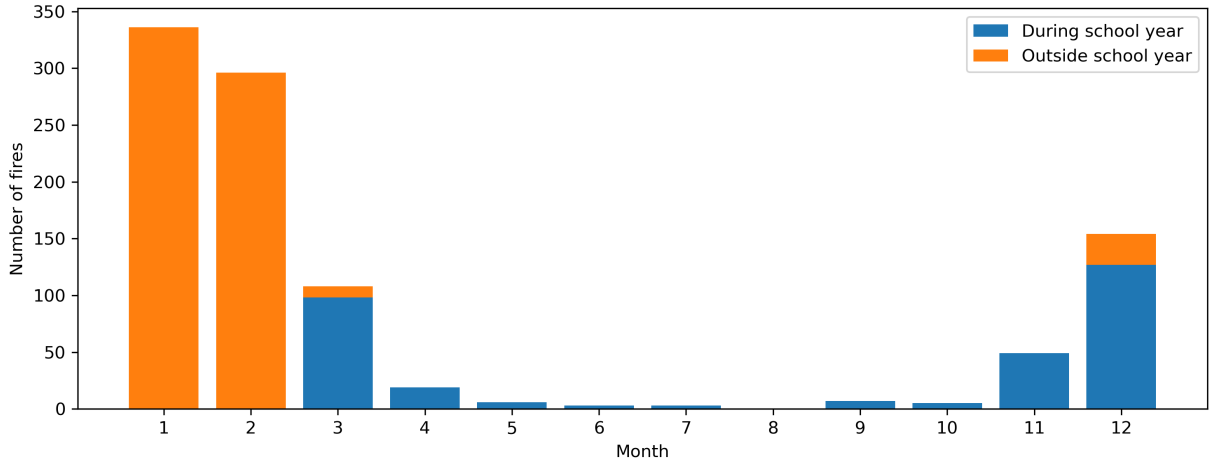


Notes: Heat map present the municipal exposures to wildfire smoke during the 2016–2017 season as modelled by HYSPLIT. A municipality is considered exposed when the particle plume, injected at 500m above the fire origin, lies within a 500km horizontal radius of the municipality, passes within 5km of its centroid, and remains at or below 100m altitude.

Figure A3: Wildfires and the school calendar



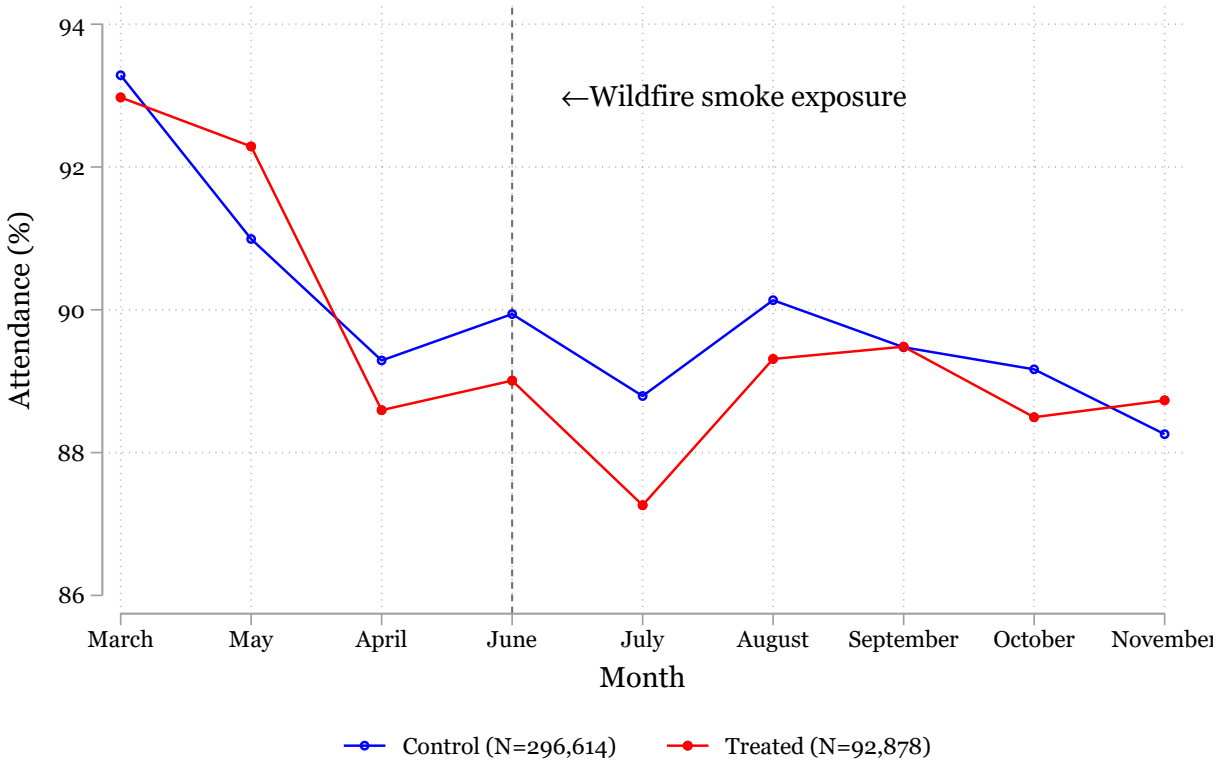
(a) Date of fire initiation and school calendar



(b) Monthly fires and school calendar

Notes: Wildfire ignition dates in Chile, 2002–2022 (CONAF). Panel (a) shows ignitions by day of year; the shaded area marks the school-year window (first Monday of March to the Sunday of the third week of December; shown approximately). Panel (b) reports monthly counts, split by whether ignitions occur during versus outside the school year. Fires with total burned area $\geq$ 200 ha are included when burned-area information is available. *Source:* CONAF. *Authors' calculations.*

Figure A4: Students' Attendance Relative to Wildfire Exposure



Notes: The sample includes students across all grades in 2013 from regions V and VII. These parameters were selected because they represent one of the few instances where wildfire exposure occurred during the school year. Consequently, the analysis compares treated and non-treated municipalities within these treated regions.